

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2022-P2-Model)** **Date: 29-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-28** **Max. Marks: 180**

29-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2022-P2)_GTA-28_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
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JEE-ADVANCE-2022-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Questions with Single Correct Choice	+3	-1	4	12
Total				**18**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

1. Let A(a) and B (b) be two distinct points on the unit circle. If Q (w) is the foot of perpendicular from point P (z) to the line joining A and B, then $\frac{4w}{a+b+z-ab\bar{z}}$ is

2. Let α_i be the solution of equation $\sin 2x = e^{-x}, \alpha_i \in [0, 2\pi]$ & i = 1, 2, 3, 4. Also $f(x) = (x + \ln\sin 2\alpha_1)(x + \ln\sin 2\alpha_2)(x + \ln\sin 2\alpha_3)(x + \ln\sin 2\alpha_4) + \tan x$. The value of $\sum_{i=1}^{4} e^{-\alpha_i}\left(f(\alpha_i) + \frac{1}{f(\alpha_i)}\right)$.

3. Let a sequence of integers $I_1, I_2, I_3, \ldots$ be defined by $I_1 = 1$ and $I_{n+1} = 1 + I_1.I_2.I_3 \ldots . I_n$ $\forall n \geq 1$. If $H_n = \sum_{k=1}^{n} \frac{1}{I_k}$ and $P_n = \prod_{k=1}^{n} \frac{1}{I_k}$ then value of $H_{2018} + P_{2018}$ is

4. Compute $\int_0^{\pi} \frac{2\sin\theta + 3\cos\theta - 3}{13\cos\theta - 5} d\theta = \frac{a\pi}{13} - \frac{b}{13}\log\frac{3}{2}$ value of a + b is _____ $(a, b \in N)$

5. Let $f(x) = \begin{cases} |x|^{p-2} \sin\left(\frac{1}{x}\right) + x|\tan x|^{q-3}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ is differentiable at x = 0, then minimum value of [p + q] equals (where [.] represents G.I.F)

6. f(x) is a nonconstant polynomial. Given that $f(f(x)) + f(x) = f(x)^2$, value of f(3) is.

7. f is bijective function from the set $\{0,1,2,...,11\}$ to $\{0,1,2,\ldots,11\}$, with the property that whenever a divides b, f(a) divides f(b). Number of such f is _____

8. Number of real zeroes of a polynomial $f(x) = x(x-2)(x-4)(x-6) + (x-1)(x-3)(x-5)(x-7)$ is _____

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

9. Let the point A, B, C have position vectors $\bar{a} = \bar{i} - 2\bar{j} + 3\bar{k}, \bar{b} = 2\bar{i} - \bar{j}, \bar{c} = 4\bar{i} + \bar{j} - 3\bar{k}$ respectively. Then

A) Area of ΔABC is $\frac{3}{\sqrt{2}}$ sq. units

B) The vectors $\bar{a}, \bar{b}, \bar{c}$ are coplanar

C) The length of the altitude through A of ΔABC is $\frac{3\sqrt{2}}{17}$

D) $\bar{a}, \bar{b}, \bar{c}$ forms a right-handed system of vectors.

10. Which of the following is/are CORRECT?

A) Let A = {2, 3, 4, 5, …..30} and '$\simeq$'be an equivalence relation on A x A, defined by (a, b) $\simeq$ (c, d), if and only if ad = bc. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair (4, 3) is 7

B) If R = {(x, y):x, y $\in Z, x^2 + 3y^2 \leq 8$} is a relation on the set of integers Z, then the domain of R^{-1} is {-1, 0,1}

C) Set A has m elements and set B has n elements. If the total number of subsets of A is 112 more than the total number of subsets of B, then the value of m.n is 28

D) Let $X = \{n \in N : 1 \leq n \leq 50\}$. If

A = $\{n \in X$:n is multiple of 2$\}$and

B = $\{n \in X$:n is a multiple of 7$\}$, then the number of elements in the smallest subset of X containing both A and B is 29.

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11. The common tangents to the circles $x^2+y^2+6x=0$ and $x^2+y^2-2x=0$ forms a triangle with centroid G, circum-centre S and in-centre I. Then

A) G = (1, 0) **B)** S = (1, 0) **C)** G = (2, 0) **D)** I = (2, 0)

12. If $z_1=a+ib$ and $z_2=c+id$ are complex numbers such that $|z_1|=|z_2|=1$ and $\mathrm{Re}(z_1\bar{z}_2)=0$, then the pair of complex numbers $\omega_1=a+ic$ and $\omega_2=b+id$ satisfy which of the following relations?

A) $|\omega_1|=1$ **B)** $|\omega_2|=1$ **C)** $\mathrm{Re}(\omega_1\bar{\omega}_2)=0$ **D)** $\mathrm{Im}(\omega_1\bar{\omega}_2)=0$

13. If function $f:R\to R$ is defined by $f(x)=|x|([x]-\sin x)$, [.] is greatest integer function then which of the following statement is/are true?

A) f is one – one **B)** f is onto

C) f is not one – one **D)** f is neither one – one nor onto

14. At a point P on the parabola $y^2=4ax(a>0)$ tangent and normal are drawn. Tangent intersects the x-axis at Q and normal intersects the parabola at R such that chord PR subtends 90° at its vertex. Then which of the following is/are TRUE?

A) PQ = $2a\sqrt{6}$ **B)** $PR=6a\sqrt{3}$

C) Area of $\Delta PQR=18\sqrt{2}a^2$ **D)** $PQ=3a\sqrt{2}$

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

15. If the sum of the real roots x to each of the equations

$$2^{2x}-2^{x+1}+1-\frac{1}{k^2}=0$$

for k = 2, 3, …, 2023 is N, value of 2^N is

A) $\frac{2022}{2023}$ **B)** $\frac{2012}{2023}$ **C)** $\frac{2023}{1012}$ **D)** $\frac{1012}{2023}$

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16. Let R be the region in the first quadrant bounded by the x-axis, the line $2y = x$, and the ellipse $\frac{x^2}{9} + y^2 = 1$. Let R' be region in the first quadrant bounded by the y-axis, the line $y = mx$, and the ellipse $\frac{x^2}{9} + y^2 = 1$, then the value of m such that R and R' have the same area is

A) $\frac{2}{9}$ **B)** $\frac{3}{11}$ **C)** $\frac{2}{5}$ **D)** $\frac{2}{7}$

17. Let f(x) be a continuous & differentiable function on [a, b].

If $g(x) = \int_a^x f(t)dt + 4\int_{b^{1/4}}^{x^{1/4}} f(t^4)t^3 dt$, and according to L.M.V.T. there exists a number c in (a,b) such that $g'(c) = \frac{\mu\left(\int_a^b f(t)dt\right)}{(b-a)}$, then μ is

A) 1 **B)** 2 **C)** 3 **D)** 4

18. Let tangents be drawn from any point P on directrix of parabola S : $y^2 = 4x$ to S intersecting it at points A and B. Locus of circumcentre of ΔPAB is

A) $y^2 = 2x - 2$ **B)** $y^2 = x - 1$ **C)** $y^2 = 4x - 4$ **D)** none of these

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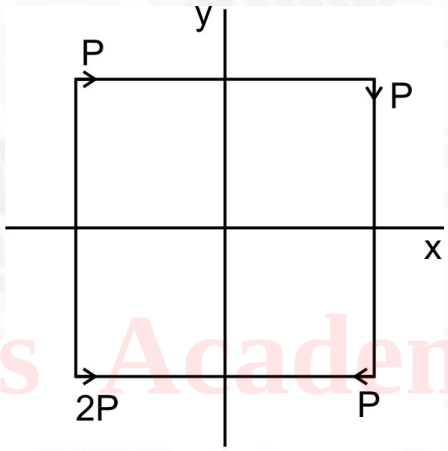
PHYSICS **Max Marks: 60**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
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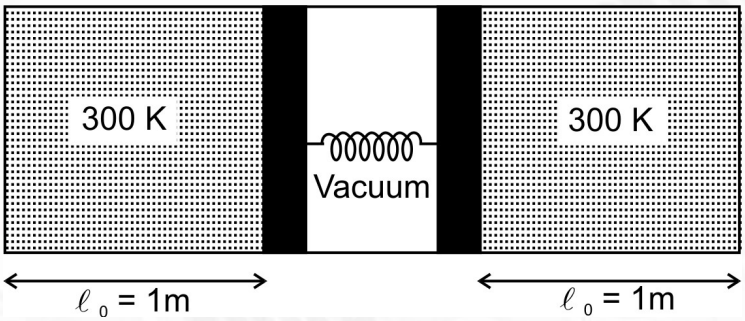
19. Four short electric dipoles, 3 each of dipole moment P and the fourth of dipole moment 2P are fixed at the vertices of an square of side length $\sqrt{2}r$ The dipole moments are oriented as shown in diagram and the centre of the square lies at the origin. The electric field strength vector at the origin is given as $\frac{KP}{r^3}(x\hat{i}+y\hat{j})$, then find value of (x + y).



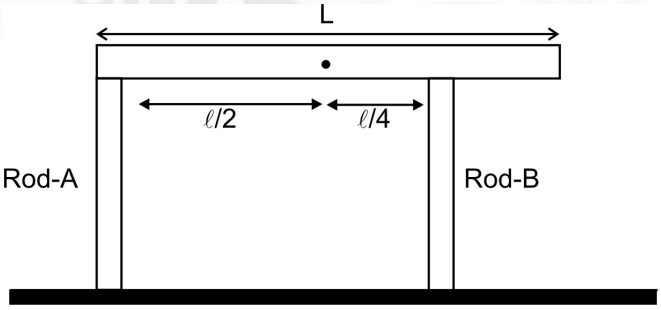
20. An inductor (L), a resistance (R) and an ideal battery (emf ε) are connected in series. Circuit is switched on at t = 0. The ratio of energy stored in the inductor at time t = L/R and energy dissipated in resistance up to that instant is $\frac{U_1}{U_2}=\frac{1}{\alpha}\frac{\left(1-\frac{1}{e}\right)^2}{\left(\frac{\beta}{e}-\frac{1}{\gamma e^2}-\frac{1}{\delta}\right)}$. Here α,β,γ and δ are integers, e is base of natural logarithm. Find $(\alpha+\beta+\gamma+\delta)$.

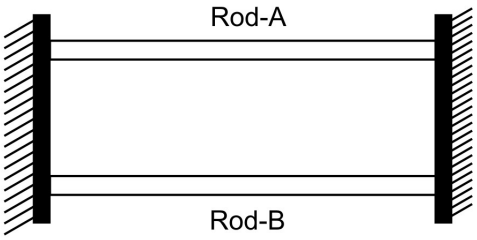
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21. Consider the shown diagram where the two chambers separated by piston-spring arrangement contain equal amounts of same ideal gas. Initially when the temperatures of the gas in both the chambers are kept at 300 K, the compression in the spring is 10 cm. The temperature of the left and the right chambers are now raised slowly to 400 K and 500 K respectively. The pistons are free to slide (i.e. no friction between piston and chamber). Final compression in the spring is found to be $\frac{\alpha}{60}$ m. Find α. (Use $\sqrt{481}=22$)



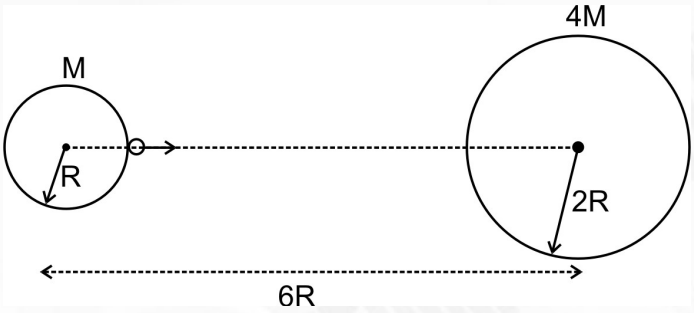
22. A uniform heavy plank of length L is resting on two rods A and B of identical length and same area of cross section but made up of different materials. It is observed that to keep the plank horizontal, one rod is to be at the end of the plank (say rod A) and another rod (say rod B) is to be at separation $\frac{L}{4}$ from centre of plank as shown in figure. Now the rods are kept between two rigid walls such that separation between the walls is equal to the natural length of the rod. Now temperature of both the rods is gradually increased in same manner, it is observed that thermal stress developed in rod A is always thrice the thermal stress developed in rod B. Ratio of coefficient of linear expansion of rod A to coefficient of linear expansion of rod B will be :



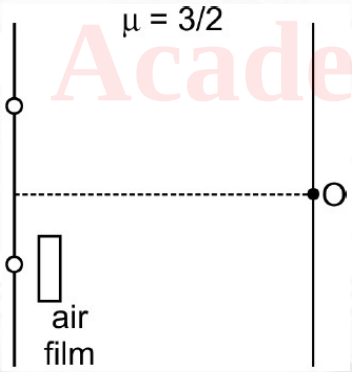


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23. Two planets of mass M & 4M are shown in the figure. Radius of the planets are R and 2R respectively and distance between centres of the planets in 6R. A small particle is thrown from planet of mass M along the line joining their centres. Minimum speed of the particle so that it can reach other planet is given by $\sqrt{\frac{x \cdot GM}{5R}}$. Find x.



24. YDSE is carried out in a liquid of refractive index $\mu = 3/2$. A thin film of air is formed in front of the lower slit as shown in the figure. A maxima is formed at point O which is mid point of screen. If thickness of air film is 4 μm then find order of maxima at point O. Wavelength of light in air is $4000\,\mathring{A}$



25. In a series LCR circuit, half power frequencies (frequency at which power disspited is half the maximum power disspited in the circuit) are found to be 90 rad/s and 40 rad/s. Find the value of $\frac{\omega_0}{10}$, where ω_0 is the resonance frequency.

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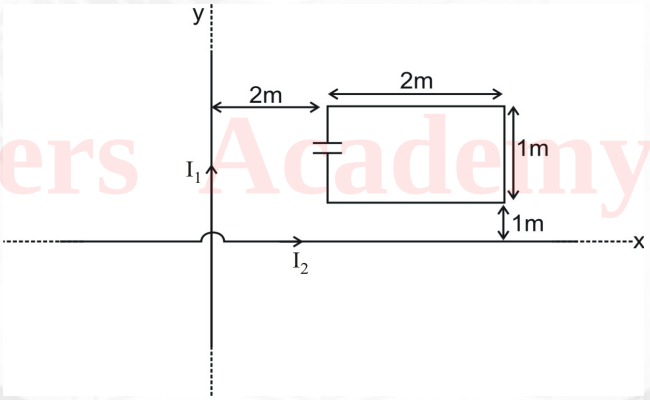

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26. A boy runs on a circular path of radius R = 28 m with a constant speed u = 4 m/s. Another boy starts from the centre of the path to catch the first boy. The second boy always remains on the radius connecting the centre of the circle and the first boy and maintains magnitude of his velocity constant at V = 4 m/s. If the time of chase is (10 + x) sec then x = ….., $\left(\pi = \frac{22}{7}\right)$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

27. Two infinite current carrying wires lie overlapping each other along x & y axes carrying currents $I_1 = 2I_0 \sin\left(\omega t + \frac{\pi}{4}\right)$ and $I_2 = I_0 \cos\left(\omega t + \frac{\pi}{4}\right)$. A rectangular frame of negligible resistance with capacitor of capacitance C is placed in the xy plane as shown. Select correct options.



A) Maximum emf induced in frame is $\frac{\mu_0 I_0 \omega \ell n 2}{2\pi}$

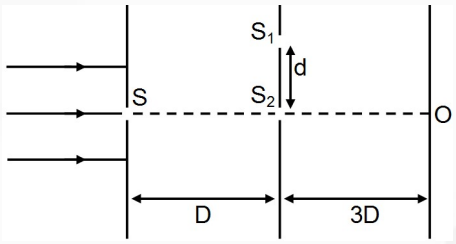
B) Maximum emf induced in frame is $\frac{\sqrt{2}\mu_0 I_0 \omega \ell n 2}{\pi}$

C) Maximum charge on capacitor is $\frac{\sqrt{2}\mu_0 I_0 \omega C \ell n 2}{\pi}$

D) Current in frame at t = $\frac{\pi}{\omega}$ is zero

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28. In the YDSE setup shown in figure a monochromatic light of wavelength λ is incident on slit S. Distance D is very large as compared to separation d between slits S_1 and S_2. Choose the correct alternative (s)



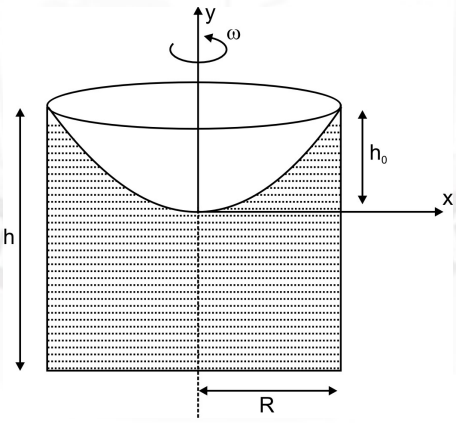
A) Minimum value of d for which there is a bright fringe at O is $\sqrt{\frac{2\lambda D}{3}}$

B) Minimumn value of d for which there is a bright fringe at O is $\sqrt{\frac{3\lambda D}{2}}$

C) Position of zeroth order bright fringe for this minimum value of d is $\sqrt{\frac{2\lambda D}{3}}$ above O

D) Position of zeroth order bright fringe for this minimum value of d is $\sqrt{6\lambda D}$ above O

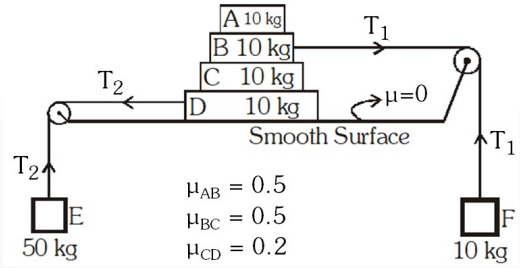
29. An ideal liquid of density ρ is filled in a cylindrical container upto (3/4)th height. Now the liquid is being rotated about vertical axis passing through its axis of symmetry with constant angular velocity ω, such that liquid is on the verge of falling out of the container and free surface of liquid forms a paraboloid. Whole situation is shown in the figure. Choose the correct option(s). [P=pressure]



A) $h_0 = \frac{h}{4}$ **B)** $\omega = \sqrt{\frac{gh}{R^2}}$ **C)** $\left|\frac{dP}{dx}\right| = \rho\omega^2 x$ **D)** $\left|\frac{dP}{dy}\right| = \rho g$

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30. Consider all pulleys and string to be ideal.



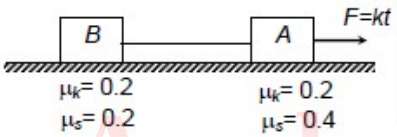
A) The value of T_2 is 200 N

B) The value of T_2 is $\frac{400}{3}$ N

C) the force of friction between B and C is 70 N

D) the force of friction between B and C is 90 N

31. Two blocks A and B each of mass $\frac{1}{2}$ kg is connected by a massless inextensible string and kept on horizontal surface. Coefficient of friction between block and surface is shown in figure. A force F = kt (where k = 1 N/s and t is time in second) is applied on A. Then ($g = 10\ m/s^2$)



A) work done by frictional force on block B is zero in time interval t = 0 to t = 3s.

B) work done by frictional force on block A is zero in time interval t = 0 to t = 3s.

C) work done by tension on B is also zero in time interval t = 0 to t = 3s.

D) speed of blocks at t = 10s is 27.5 m/s.

32. A particle starts SHM at time t = 0. Its amplitude is A and angular frequency is ω. At time t = 0 its kinetic energy is $\frac{E}{4}$, where E is total energy. Assuming potential energy to be zero at mean position, the displacement – time equation of the particle can be written as

A) $x = A\cos\left(\omega t + \frac{\pi}{6}\right)$ **B)** $x = A\sin\left(\omega t + \frac{\pi}{3}\right)$

C) $x = A\sin\left(\omega t - \frac{2\pi}{3}\right)$ **D)** $x = A\cos\left(\omega t - \frac{\pi}{6}\right)$

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JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
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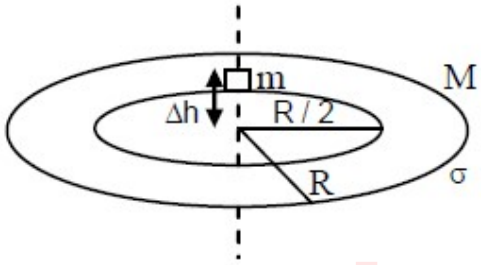
SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

33. A small block of mass m lies above a thin disk of total mass M, constant surface density σ, and radius R. A hole of radius R/2 is cut into the center of the disk, allowing the mass to pass though. The mass starts at a position along the central axis of the disk but is displaced a height Δh from the plane of the disk, where $\Delta h << R$. There are no other external forces at work on this system; i.e., the disk and block are out in deep space. Assume m << M. For the first order in Δh the block undergoes simple harmonic motion with respect to the centre of the plane of the disk, compute the period of oscillation of block.



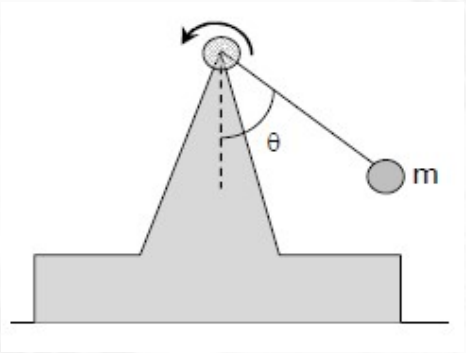
A) $T \simeq \sqrt{\frac{2\pi R}{G\sigma}}$ **B)** $T \simeq \sqrt{\frac{\pi R}{G\sigma}}$ **C)** $T = \sqrt{\frac{\pi R}{2G\sigma}}$ **D)** None of these

34. Consider a metal can placed coaxially inside a long solenoid. The metal can be modelled as a thin cylindrical shell of radius r. The magnetic field due to the solenoid varies with time t as kt where k is a positive constant. The resistivity of the can is ρ and its breaking stress is σ. Ignore the self –inductance of the can and any currents induced in the flat surfaces of the can. Assume that the magnetic field produced by the induced currents is negligible. Compute the time at which the can breaks.

A) $t = \frac{\sigma\rho}{r^2K^2}$ **B)** $t = \frac{2\sigma\rho}{r^2K^2}$ **C)** $t = \frac{\sigma\rho}{2r^2K^2}$ **D)** $t = \frac{\sigma\rho}{3r^2K^2}$

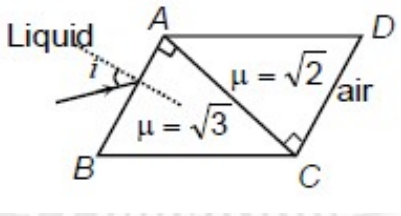
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35. A motor is installed at the top of a pole rigidly fixed on a platform. A light rod of length r = 1 m is rigidly attached to the motor shaft at its one end and at the other end a small ball of mass m is attached. The rod can rotate in a vertical plane with the help of the motor. Total mass of the platform, the pole and the motor is $\eta = 4$ times the mass of the ball. The motor rotates the rod at a constant angular velocity. The platform is placed on a horizontal surface, where coefficient of friction is $\mu = 1/\sqrt{3}$. At which minimum angular velocity ω_0 of the rod, will the platform start sliding?



A) $\omega_0 = 2rad/S$ **B)** $\omega_0 = 3rad/S$ **C)** $\omega_0 = 5rad/S$ **D)** None of these

36. A ray of light from a liquid $\left(\mu = \sqrt{3}\right)$ is incident on a system of two right angled prism of refractive indices $\sqrt{3}$ and $\sqrt{2}$ as shown. The ray suffers zero deviation when emerges into air from CD. The angle of incidence i is



A) 45° **B)** 35° **C)** 20° **D)** 10°

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CHEMISTRY **Max Marks: 60**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

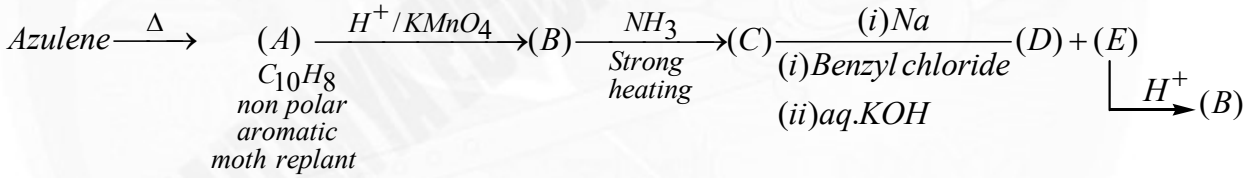
Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

37. (I) When chromite ore is fused with Na_2CO_3 in free excess of air, the mole ratio of combination of chromite ore, Na_2CO_3 and O_2 is $x_1 : x_2 : x_3$ respectively in formation of sodium chromate (x_1, x_2 and x_3 are the smallest integer in balance reaction)

(II) Atomic number of 3d element with highest $E^{o}_{M^{2+}/M}$ (except Sc) is x_4,

Then find the value of $\left[\frac{x_4+1}{x_1+x_2-x_3}\right]$.

38.



(Given: B is a structural isomer of Terephthalic acid)
If number of contributing resonating structure of A are x and degree of unsaturation of D is y, then x + y is?

39. 1mole of a polypeptide of hydrolysis produced 4 moles of glycine, 2 moles of alanine, 1 mole of phenyl alanine and 1 mole of leucine.
If x = number of peptide linkages in polypeptide molecule
y = possible number of polypeptide if all identical amino acids remain together in all polypeptides.
z = Mass of water (gm) gained by 1 mole of polypeptide on complete hydrolysis.
Calculate : $\frac{yz}{108x}$

40. All possible carbonyl compounds with molecular formula (C_8H_8O) which has a benzene ring in its structure and gives positive Tollen's test, when reacted with hydroxylamine how many organic products will be possible with difference in melting point?

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41. $NaBH_4 + I_2 \rightarrow Products$

How many total moles of gaseous products that are obtained using 6 moles of $NaBH_4$ and 3 moles I_2

42. $[Au(CN)_2]^-$ is a very stable ion and formation constant of $[Au(CN)_2]^-$ is 4×10^{28}.

$x\times10^{-y}$ M concentration of CN^- ion is required to maintained equilibrium such that 99% (by mole) of Au^+ is in form of $[Au(CN)_2]^-$, Find the value of x + y

43. Concentration of H_2SO_4 and Na_2SO_4 in a solution is 1 M and 1.8×10^{-2} M, respectively. Molar solubility of $PbSO_4$ in the same solution is $X\times10^{-Y}$ M (expressed in scientific notation). The value of Y is _____.

[Given: Solubility product of $PbSO_4(K_{sp}) = 1.6\times10^{-8}$. For H_2SO_4, K_{al} is very large and $K_{a2} = 1.2\times10^{-2}$]

44. Find out total number of paramagnetic inner orbital complex

$K_2[NiCl_4], K_3[Fe(CN)_6], K_3[Cr(CN)_6], K_2[PtCl_4]$, $Na_2[Fe(CN)_5NO]$, $[Fe(H_2O)_5NO]SO_4$, $K_3[Co(C_2O_4)_3], K_4[Co(CN)_6]$, $[Cu(H_2O)_6]SO_4$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

45. Which of the following statement(s)/reactions is/are correct?

A) Co^{3+} is a good oxidising agent

B) $Cu(s) + ConcH_2SO_4 \rightarrow CuSO_4(aq) + H_2O + SO_2$

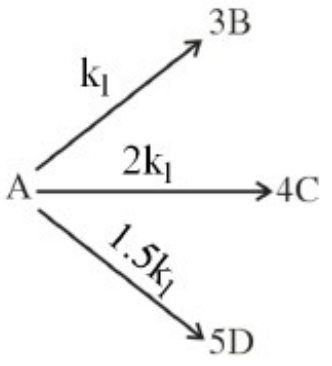
C) Cr_2O_3 and V_2O_5 are amphoteric oxides while Mn_2O_7 is acidic oxide

D) $2CrCl_2(s) + 2HCl(dil.) \rightarrow 2CrCl_3(aq) + H_2(g)$

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46.



All reactions are 1st order.

At time t = t_1 (t > 0), $\frac{[B]}{[C]} = \alpha$ and at time t = t_2 (where $t_2 \geq t_1$, $\frac{[C]}{[D]} = \beta$

Which of the following is/are correct?

A) $\alpha = 0.5$ **B)** $\beta = 4/3$ **C)** $\alpha = 0.375$ **D)** $\beta = 16/15$

47. Select the correct statement.

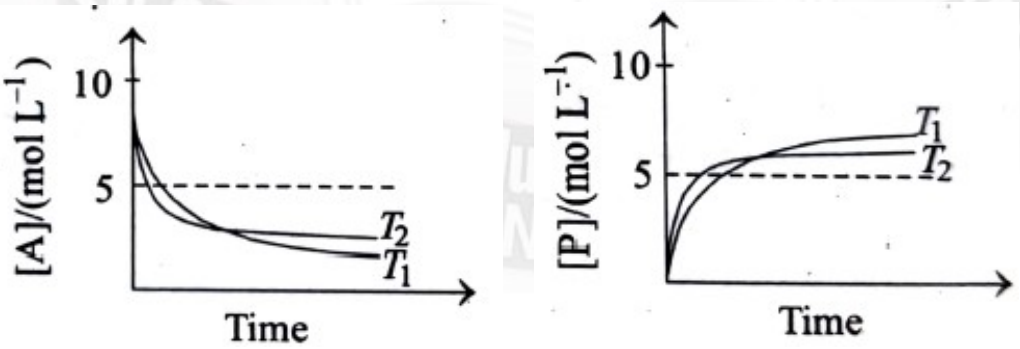
A) D-Glucose and L-Galactose are C-4 epimers

B) Sucrose does not undergo Mutarotation

C) *D – Gluocse* and L-Galactose are components of lactose

D) D-glucose upon treatment with warm HNO_3 forms D-glucaric acid.

48. For a reaction, $A \rightleftharpoons P$, the plots of [A] and [P] with time at temperatures T_1 and T_2 are given below.



If $T_2 > T_1$, the correct statement(s) is (are)

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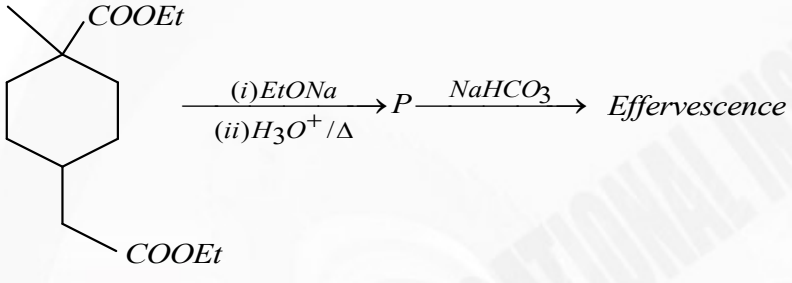
(Assume ΔH^o and ΔS^o are independent of temperature and ratio of lnK at T_1 to lnk T_2 is greater than $\frac{T_2}{T_1}$. Here H, S, G and K are enthalpy, entropy, Gibbs energy and equilibrium constant, respectively.)

A) $\Delta H^o < 0, \Delta S^o < 0$ **B)** $\Delta G^o < 0, \Delta H^o < 0$

C) $\Delta G^o < 0, \Delta S^o < 0$ **D)** $\Delta G^o < 0, \Delta S^o > 0$

49. Which of the following observation is/are correct?

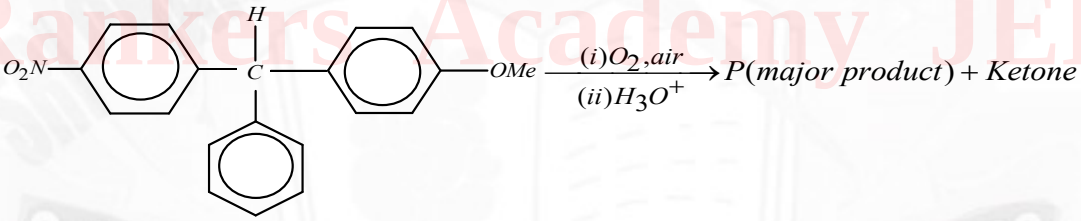
A)



B) $C_6H_{14} \xrightarrow[(ii)Oleum\ (iii)NaOH/\Delta\ (iv)H_3O^+]{(i)Al_2O_3-Cr_2O_3,\Delta} A \xrightarrow[conc.H_2SO_4]{Phthalic\ Anhydride} \xrightarrow{NaOH} Pink\ colour$

C) $\underset{(Alcohol)}{C_2H_6O} \xrightarrow[(ii)Et_2SO_4]{(i)Base} \xrightarrow{Air/h\upsilon} \xrightarrow[KCNS]{FeSO_4} Red\ colour$

D)



$P + NaHCO_3 \rightarrow Effervescence$

50. The compound(s) formed upon combustion of sodium metal in excess air is (are)

A) Na_2O_2 **B)** Na_2O **C)** NaO_2 **D)** NaOH

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

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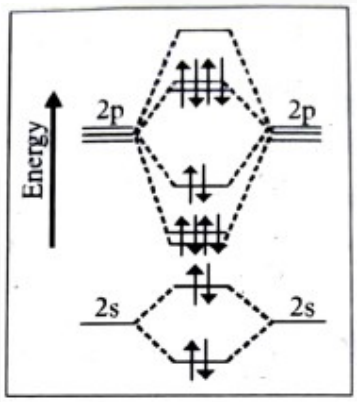
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51. The IUPAC nomenclature of an element with electronic configuration $[Rn]5f^{14}6d^{1}7s^{2}$ is:

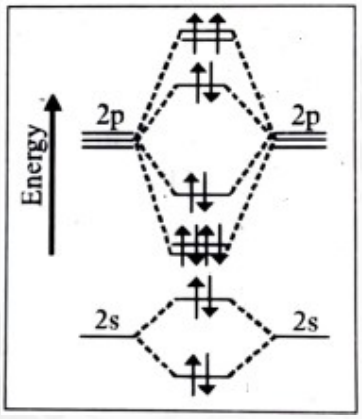
A) Unnilbium **B)** Unnilunium **C)** Unnilquadium **D)** Unniltrium

52. The correct molecular orbital diagram for F_2 molecule in the ground state is

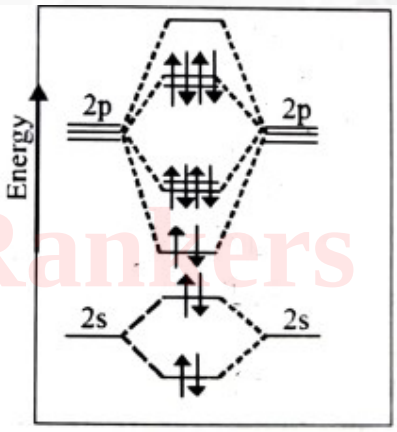
A)



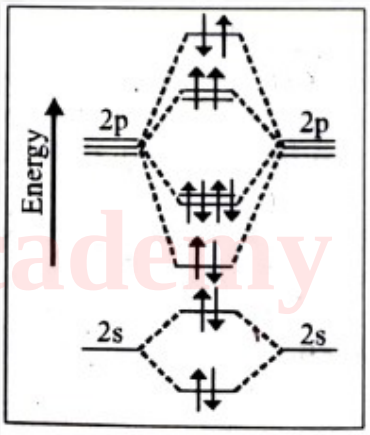
B)



C)



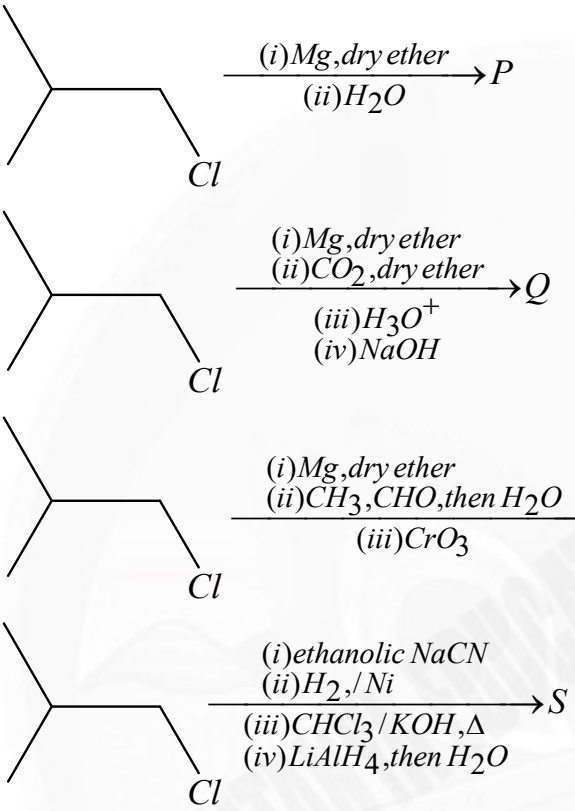
D)



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53. In the following reactions P, Q, R, and S are the major products.



The correct statement about P, Q, R and S is

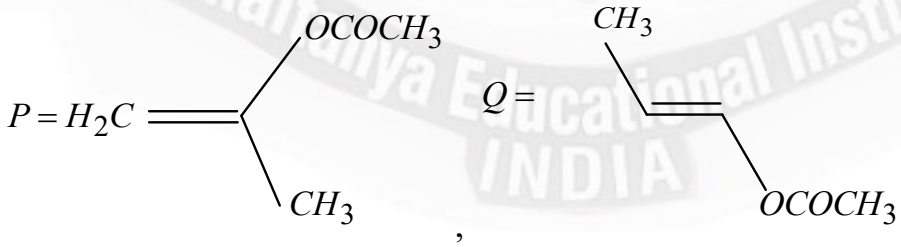
A) P is a primary alcohol with four carbons.

B) Q undergoes Kolbe's electrolysis to give an eight-carbon product.

C) R has six carbons and it undergoes Cannizzaro reaction.

D) S is primary amine with six carbons.

54. The product of acid hydrolysis of P and Q can be distinguished by



A) Lucas Reagent

B) 2, 4- DNP

C) Fehling's Solution

D) $NaHSO_3$

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